Planck and Bohm the Men Whose Fate Is “Uncomputed”

—— Part 4: The Three Sacred Treasures ——

Characters:

- **The Objective Observer (Salviati):** One who possesses deep insights into Digital Signal Theory and the discrete universe.
- **The Existing Intellect (Simplicio):** One who reveres Quantum Mechanics as a "sublime truth."
- **The Seeker (Sagredo):** One who seeks to unravel the universe's specifications through simple logic.

Act I: The Superordinate Concept of Real Numbers

Sagredo: Quantum mechanics, represented by the Schrödinger equation, is naturally a superordinate concept to Newtonian mechanics—including even Special Relativity. And the decisive difference lies in the fact that while Newtonian mechanics is a mechanics of real numbers, quantum mechanics is a mechanics of complex numbers.

Act II: Tonomura’s Great Experiment

Simplicio: So, it comes down to whether or not one possesses the concept of "phase information," which cannot be described by real numbers alone.

Salviati: Precisely. This phase information is the fundamental concept that forms the core of quantum mechanics. The definitive proof of this is the verification experiment of the Aharonov-Bohm (AB) effect conducted by Akira Tonomura.

Simplicio: He confirmed that in a region where no magnetic field (B) exists, the vector potential (A) exerts an influence on the phase information of the electron’s matter wave. That was a monumental experiment that established A not as a fictional product of mathematical technique, but as something "real and more fundamental."

Act III: The Three Sacred Treasures

Salviati: Correct. However, the fact that A is fundamental and real is something Maxwell had predicted from the beginning. For us today, having already assimilated Tonomura’s experimental results, the greatest value of this experiment lies in accessing a concept that does not emerge from Maxwell’s electromagnetism—namely, the "phase information of the electron"—and visualizing the "digital transition" of that phase.

Sagredo: A "toroidal magnet," a "superconductor," and a "coherent electron gun." So, with these **Three Sacred Treasures**, the moment of phase computation was finally brought to light?

Act IV: A Natural Conclusion

Salviati: Exactly. In classical mechanics, E and B—the sources of "force" (the absolute value of complex numbers)—are the protagonists. But the moment we enter the realm of quantum mechanics, A takes the leading role. And it should; for A was the only thing capable of influencing the phase.

(Curtain Falls)